



Development of a mechanically robust silicon-based cross-linking polymer for the sustainable marine antifouling coatings

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ABSTRACT

Silicone-based fouling release coatings (FRCs) have shown great promise as an environmentally friendly anti-fouling technology, while inferior mechanical properties and insufficient substrate adhesion have hindered their further application. We focused on enhancing the mechanical robustness of silicone-based coating through the synergistic effects of hydrogen bonding, π - π aromatic interaction and covalent bonding. The resulting silicone-based coating displayed a low surface energy of 23 mJ/m² and improved mechanical strength and substrate adhesion, offering values up to 6.4 MPa (9.1 times higher) and 2.87 MPa (25.7 times higher), respectively. Additionally, the modified coating demonstrated a tear strength of 9.8 kN/m and mass loss only about 30 mg after enduring 10,000 wear cycles. Alkoxysilane-functionalized polyethylene glycol and quaternary ammonium salt were grafted into the cross-linking structure to make them anchored to the networks and migrated to the surface layer in seawater without releasing. The coating provided 99.1% antibacterial efficiency and effective antidiatom performance (≤ 11.2 cell/mm²). After a 12-month marine field test, the developed film showed only 20% micro bio-fouling compared to 100% fouling coverage on the PDMS surface. By combining improved mechanical properties with antifouling characteristics, this new multifunctional silicone-based FRC exhibits tremendous potential for sustainable marine antifouling coating application.

1. Introduction

The growth and accumulation of diverse marine organisms on ship hulls and submerged structures, including microorganisms, barnacles and seaweeds, commonly referred to as marine biofouling, has been a persistent issue with detrimental effects on both the economy and the environment [1]. Marine biofouling increases the roughness of ship hulls, leading to higher frictional resistance, increased fuel consumption, and additional operational costs. It was reported that 5% barnacle fouling could increase drag resistance by up to 66%, which would require extra power for propulsion [2]. In response, antifouling coatings have been developed since the mid-20th century, which is the most convenient and effective method to prevent marine biofouling. Generally, traditional antifouling coatings include toxic biocides, which can potentially harm the ecosystem. Ecofriendly antifouling coatings, such as antifoulant release coatings (ARCs) [3,4], fouling resistance coatings

[5,6], fouling release coatings (FRCs) [7,8] and biomimetic antifouling surface [9–11], have been explored to inhibit biofouling. Among these strategies, fouling releasing coatings (FRCs) have attracted considerable attention due to their ability to weaken the adhesion strength of fouling organisms [12,13], resulting in effective antifouling performance.

The poly(dimethylsiloxane) (PDMS) elastomer is widely employed as the primary material for FRCs, owing to its low surface energy (18 mJ/m²) [14], low modulus (0.013 to 1.4 MPa) [15] and low surface roughness (4 to 278 nm) [16]. Despite the advantages of their eco-friendly nature, drag reduction and chemical stability of silicone-based FRCs, their applications are still restricted by various drawbacks. One major drawback is the inherent poor mechanical property, which render them susceptible to mechanical damage (impacting, cutting, tearing and puncturing), thereby diminishing their longevity [17–20]. Moreover, the natural low adhesion of FRCs to substrates mandates the requirement of the tie coat to amplify the adhesion between the FRCs and the

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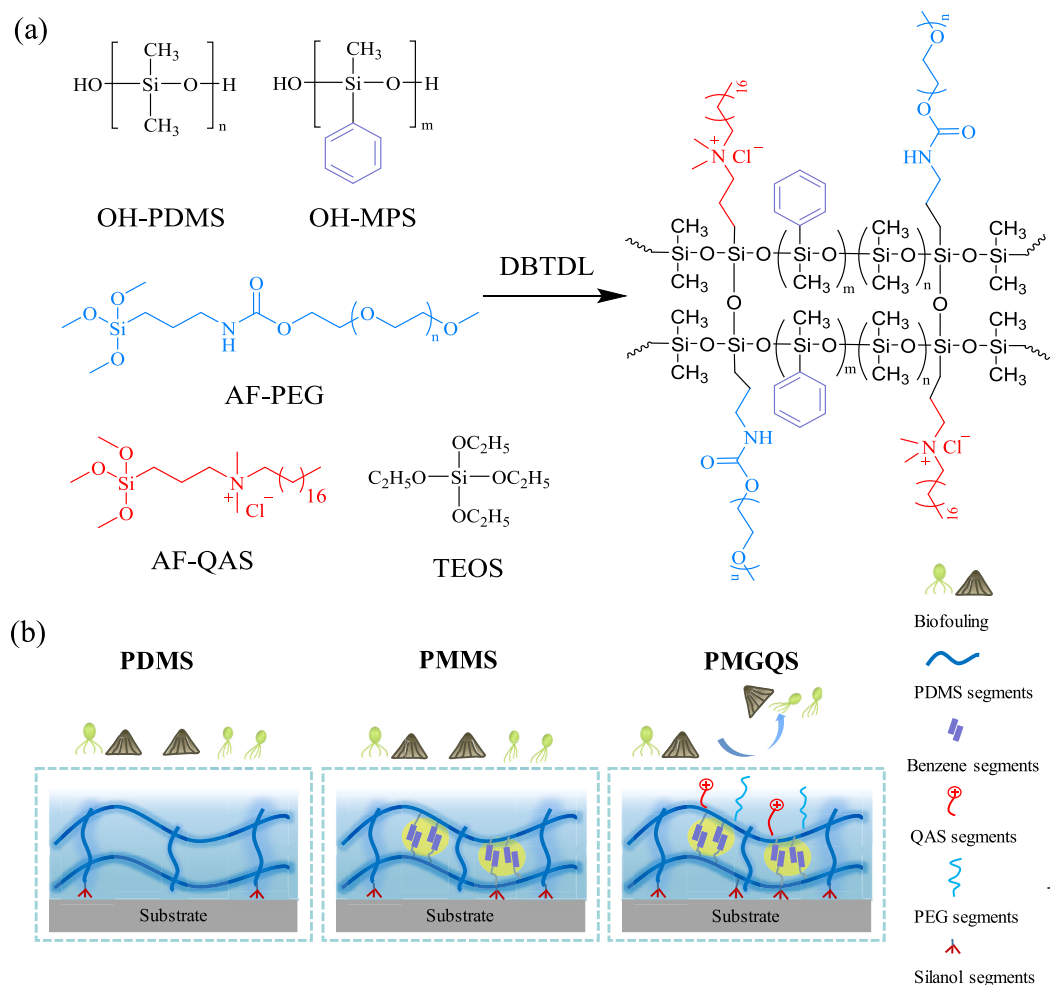
primer paint. This subsequently leads to a rise in costs and production time, potentially restricting more applications [7,21]. Another important issue is that silicone elastomers can not possess the capability to impede the buildup of diatom and bacterial slime on stationary marine structures or during ship idleness [22,23], rendering its removal arduous even when the vessel is traveling at a velocity of 30 knots or with the assistance of a water jet [24]. Consequently, there is a pressing need to enhance the fouling resistance of silicone-based FRCs, particularly under static circumstances.

To overcome these limitations, both physical and chemical modifications have been employed. The mechanical strength and adhesion capabilities of silicone-based FRCs can be improved through physical integration with nanofillers, including carbon nanotubes (CNTs) [25], graphene oxide (GO) [26–29], aluminum nanorods and titanium oxide (TiO₂) nanospheres [30]. Besides that, researchers have investigated the inclusion of urethane [31,32], urea [33] and epoxy groups into PDMS to boost its mechanical strength [34]. Nevertheless, these approaches compromise the antifouling properties of PDMS due to the surface changes induced by these polar groups. In the past decades, significant progress has been made in the research of bioinspired chemical design adhesion mechanisms from marine organisms [35–37]. Hydrogen bonding, π - π aromatic interaction and covalent bonding complexation play the important role in curing and interfacial interactions [36,38]. However, there are few reports on chemical design FRCs with strong interfacial adhesion to substrate keeping low surface energy [8,31].

Additionally, coating surfaces with chemically immobilized biocidal

moieties such as organic antifoulants, quaternary ammonium salts (QAS), amphiphiles and zwitterions to avoid their release into the ocean during immersion is a novel and excellent approach to render the surface antifouling [5,9,39]. The main focus in terms of biocidal compounds have been on quaternary ammonium salts, which are known for their ability to kill microorganisms by interacting with their cell walls through electrostatic and lipophilic interactions [40]. Studies have demonstrated that surfaces coated with QAS-containing polymers are effective at killing microorganisms, such as gram-positive bacteria, gram-negative bacteria, yeasts and molds [41]. These multifunctional synergistic strategies potentially enhanced the antifouling capabilities and lifespan of the coatings.

This investigation aims to improve the inferior mechanical properties and insufficient substrate adhesion that hinder applications involving mechanical abrasion and hard impacts. Additionally, it seeks to address the challenges of fouling protection at low sailing speeds or under still conditions, which currently require the release of biocides into the sea, causing ecological consequences. Herein, we developed eco-friendly and high-efficiency modified silicone-based FRC with robust mechanical properties and sustainable marine antifouling performance. Utilizing bis-silanol-terminated methyl phenyl silicone (OH-MPS), the modified silicone-based coating demonstrated formidable mechanical strength and interfacial adhesion, attributable to hydrogen bond, π - π aromatic interaction and covalent bonding complexation. Furthermore, the coating has low surface energy, outstanding tear strength and wear resistance, remarkable antibacterial and antidiatoms



Scheme 1. Synthesis route and responsive mechanism of mechanically robust silicone-based cross-linking polymers. (a) Molecular structures of OH-PDMS, OH-MPS, AF-PEG, AF-QAS and TEOS, and synthesis of the silicone-based cross-linking polymers. (b) Schematic illustration of antifouling and enhanced mechanical mechanism of PMGQS.

adhesion, showing satisfactory fouling release properties. We employed alkoxyisilane-functionalized polyethylene glycol (AF-PEG) and quaternary ammonium salt (AF-QAS) to integrate into the methyl phenyl silicone cross-linking structure (PMGQS) via hydrolysis and condensation reactions, providing antibacterial and antifouling performance, and antifouling compositions can be anchored on the surface to form antifoulant-riched layer instead of being released in water (Scheme 1). With its enhanced mechanical properties and antifouling characteristics, this new multifunctional silicone-based FRC would broaden the technology applications in the marine industry, not only enhancing vessel performance, reducing maintenance costs and prolonging protection life, but also mitigating ecological impacts.

2. Experimental section

2.1. Materials

OHX-4010 silicone polymer resin (OH-PDMS) was purchased from Nanjing Golden Chemical Co., Ltd. Dimethoxymethylphenylsilane (98%), tetraethyl orthosilicate (> 99%), ditin butyl dilaurate (DBTDL, 95%), 3-isocyanatopropyltrimethoxysilane (97%), methoxypolyethylene glycol (MPEG), average M_w 1900, and 3-(trimethoxysilyl) propyl dimethyl octadecyl ammonium chloride (AF-QAS) solution (60 wt% in methanol, which was removed through evaporation before material preparation) were purchased from Shanghai Aladdin Biochemical Technology Co., Ltd. Toluene ($\geq 99.5\%$), trichloromethane ($\geq 99\%$), and ethanol (99%) were purchased from Shanghai Chemical Reagents Co., Ltd. and used directly. Artificial seawater (ASW) was prepared according to ASTM D114-98 (2003). *S. aureus* (ATCC 6538) and *E. coli* (ATCC 25922) were purchased from China General Microbiological Culture Collection Center (CGMCC), *Navicula* and *Nitzschia closterium* used for the anti-biofouling test were purchased from Xia Men (China).

2.2. Synthesis of bis-silanol-terminated methyl phenyl silicone (OH-MPS)

First, dimethoxymethylphenylsilane (DMMPs, 27.3 g, 150 mmol) was dissolved in THF (300 mL), hydrochloric acid solution (67.5 g, PH = 1) was added while stirring, and the reaction was carried out at room temperature for 24 h and 60 °C for another 24 h. Then, THF was removed and the product was extracted with CHCl_3 , washed with water, and dried with anhydrous calcium chloride for 2 h. Finally, CHCl_3 was removed, and the viscous silicone precursor was obtained (150 mmol, 100%) and tested by ^1H NMR and FT-IR (Fig. S1–3).

2.3. Synthesis of alkoxyisilane-functionalized polyethylene glycol (AF-PEG)

5.0 g of methoxypolyethylene glycol (PEG) was mixed with 1.0 g of 3-isocyanatopropyltrimethoxysilane (IPTS) (1:1 M ratio) and dissolved in 25 mL of CHCl_3 . 2 drops of butyltin dilaurate (DBTDL) were added, and the mixture was heated at 65 °C for 12 h. Then, the reaction mixture was coagulated in hexane and purified. The polymer was obtained by filtration and then dried to constant weight at room temperature under vacuum (5.1 g, 92%). It was tested by ^1H NMR and FT-IR (Fig. S4–6).

2.4. Preparation of the modified silicone-based polymers

The modified silicone-based coatings were prepared by a mixture of condensation reactions involving Si-OH and alkoxyisilane. Taking PMGQS as an example, OH-PDMS (7.5 g), OH-MPS (2.5 g), AF-PEG (0.5 g), AF-QAS (0.5 g), TEOS (1.2 g) and DBTDL (0.05 g) were dissolved in toluene (10 mL) and stirred with a magnetic bar for 1 h. The mixed solution was sprayed and cured at room temperature for 5 days to ensure cure completely. The silicone-based coatings are designated PDMS, PMMS, PMGS, PMQS and PMGQS, and the compositions of the different coatings are shown in Table 1.

Table 1

Composition of modified silicone-based polymers.

Sample	OH-PDMS (g)	OH-MPS (g)	AF-PEG (g)	AF-QAS (g)	TEOS (g)	DBTDL (g)
PDMS	10	0	0	0	1.2	0.05
PMMS	7.5	2.5	0	0	1.2	0.05
PMGS	7.5	2.5	1	0	1.2	0.05
PMQS	7.5	2.5	0	1	1.2	0.05
PMGQS	7.5	2.5	0.5	0.5	1.2	0.05

2.5. Characterizations

^1H NMR spectra were recorded on a Bruker 300 MHz nuclear magnetic resonance (NMR) (Bruker, Germany) using CDCl_3 or DMSO as the solvent and tetramethylsilane (TMS) as the internal reference at room temperature (25 °C). Fourier transform infrared (FT-IR) spectra were measured on a Bruker TENSOR 27 (Bruker, Germany) FT-IR spectrometer (KBr disk). Both the number-average molecular weight (M_n , GPC) and molecular weight distribution ($\mathcal{D}$) of all the polymers were determined by using a TOSOH HLC-8320 gel permeation chromatograph (GPC) (TOSOH, Japan) equipped with a refractive-index detector (THF) using a TSK gel Multi pore HZ-N (3) 4.6×150 mm column. All the GPC samples were injected using a TOSOH plus autosampler, and the molar masses were calibrated with narrow polydispersity using polystyrene or poly(methyl methacrylate) standards followed by universal calibration. Atomic force microscopy (AFM) (Olympus, Japan) was performed with an AC 240-type tip. Laser scanning confocal microscopy (LSCM) (Olympus, Japan) images were obtained by an Olympus LEXT Nano Search Microscope (OLS4500, Japan). XPS characterization was performed using a ULVAC-PHI (PHI 5000 Versaprobe II) (Thermo Fisher Scientific, Germany) with a 650 μm diameter and a fluenceenergy of 50 eV to collect high-resolution XPS spectra. Wide-angle X-ray diffraction (WAXD) patterns were obtained using a Rigaku MiniFlex 600 X-ray diffractometer (Rigaku, Japan) with Cu $K\alpha$ radiation and a scan speed of $10^\circ \text{ min}^{-1}$. Two-dimensional small-angle X-ray scattering (2D-SAXS) patterns were conducted in SAXSpace (Anton Paar, Austria) using Cu $K\alpha$ radiation source. The image exposure time was 600 s $\times$ 3 times and the X-ray wavelength was 1.542 Å. The distance between the detector and the sample was 307.3065 mm.

2.6. Water contact angle (WCA) and surface free energy (SFE)

Static contact angle measurements were conducted on an OCA20 machine (Data-Physics, Germany) by depositing 2 μL water or diiodomethane droplets on a PDMS film surface coated on glass slide at 25 °C. The average of the results from five measurements was used as the final CA value. The surface free energy was calculated using the Owens-Wendt-Rabel-Kaelble method. [41].

2.7. Mechanical property testing

Tensile tests were carried out on an Instron Universal Testing Machine (1125IBS) with a 5 kN loading cell. The cross-head speed of the tensile measurements was 100 mm/min. To avoid systematic failure in the vicinity of the clamps, according to the GB/T2941 standard, all the samples were cut using a dumbbell-shaped die cutter (type 3), and the thickness was approximately 1 mm. The testing temperature and humidity were 25 °C and 50%, respectively. Tear strength measurements were conducted using an Instron Universal Testing Machine (1125IBS) with a 5 kN loading cell at a speed of 500 mm/min. The samples were made into right-angle shapes according to ISO34-1: 2004 for tear strength measurements [42,43]. An abrasion device (BGD 523, Guangzhou Biaogeda Precision Instrument Co., Ltd.) was used to measure the wear resistance of the coating, as shown in Fig. S7. The coating sprayed on a disc with a diameter of 10 cm was placed under the press of

a 500 g grinding wheel that was wrapped with 150-grit sandpaper. The coated disc was rotated under the grinding wheel at a speed of 60 r/min. The mass loss was measured every 1000 rotations until reaching 10,000 rotations. This abrasion method is described in ASTM-3359 and ASTM-D4060 (Standard Test Method for Abrasion Resistance of Organic Coatings by the Taber Abraser) [44,45]. The adhesion of the network-like MH layer to iron substrate was evaluated through cross hatch adhesion test, according to ASTM D335909 standard. The area for test with size of 10 mm × 10 mm was cut deeply to the iron substrate with sharp blade, and divided into 100 grids. After bushing away the MH flakes, the sample surface was well adhered with 3 M 600-HC33 transparent tape. After 90 s press, the tape was removed, and the detachment of MH layer was assessed to evaluate the adhesion according to the standard from 0B to 5B. Besides, tape peeling test was also carried out according to the standard method B of ASTM D 3359–09. The evaluation of impact resistance of the coatings applied on the panels was carried out in accordance with ISO 6272-1: 2002 standard. In this test, a spherical indenter of diameter 20 mm impacts the coated sample on the back side. The drop height is adjusted to 25–100 cm to determine the impact energy to create damage. The adherence of the coating on the top side is checked using adhesive tape.

2.8. Antibacterial performance evaluation

The antibacterial performances of the different coatings were investigated against *Escherichia coli* (*E. coli*, ATCC 25922) and *Staphylococcus aureus* (*S. aureus*, ATCC 6538) with the plate count method according to ISO 22196. Basically, the bacteria were inoculated into 10 mL normal saline (pH = 7.2–7.4) containing 2.5% nutrient broth (NB), and the initial concentration of the bacterial suspension was approximately 10^8 cfu/mL as determined by turbidimetry. The *E. coli* suspension and *S. aureus* suspension were diluted with normal saline containing 2.5% NB to 10^5 cfu/mL. 100 μ L of the diluted suspension was dripped on a polyethylene wrap film (20 mm × 20 mm), and the sample (25 mm × 25 mm) was placed on the center of the film and wrapped up by the same film. After incubation for 24 h at 37 °C, the bacterial suspension was diluted with 10 mL of soya casein digest lecithin polysorbate (SCDLP) broth and further diluted with normal saline to a countable concentration. After incubation for 24 h, the colonies were determined by plate count agar, and the assays were repeated three times. The antibacterial rate was calculated by the following equation:

$$E_b = \frac{N_b - N_c}{N_b} \times 100\%.$$

where N_b and N_c are the average number of bacteria on the blank and coated samples, respectively.

2.9. Anti-biofouling test

In this assay, the adhered algae on different substrates were visualized by autofluorescence image analysis (Olympus, Th4–200) using a halogen lamp. *Navicula* and *Nitzschia closterium* were cultured and used for settlement bioassays. Before testing, the samples were equilibrated in artificial seawater for 24 h. Then, all the samples were placed in a 6-well plate and covered with an algal suspension of *Navicula* and *Nitzschia closterium* (5 mL). Then, the 6-well plate was placed in an incubator at 23 °C for 24 h. Before the determination of the attached diatoms, all the samples were washed cautiously using ASW three times to remove any loosely bound diatoms. All data were from at least 2 independent experiments and 2 parallel samples, and each sample provided a total of 20 measurement points, each being 0.41 mm². The data are expressed as the mean standard deviation (SD), and typical images are shown. All images were analyzed using IMAGEJ software.

The antifouling performance in sea was evaluated according to the ASTM D6990–05 (2011) method by continuously immersing coated panels in seawater at Xiamen Bay, Southeast China, for twelve months (May 2021 to May 2022). The panels were removed and inspected at 3

months interval, after being photoed, panels were gently washed with seawater to remove silt and unattached forms, and immediately placed back to continue the test. The quantitative data of the area covered by biofouling were determined by IMAGEJ [46].

3. Results and discussion

3.1. Characterization of chemical structures

In this work, FT-IR, SEM and XPS were measured to investigate changes in the chemical composition of the PDMS, PMMS, PMGS, PMQS and PMGQS surfaces in the process of chemical reaction. As shown in Fig. S8, the disappearance of 3294 cm^{−1} (Si-OH stretching vibration) showed that all OH-PDMS and OH-MPS components in the system were completely consumed during the reaction. Peaks at 700 cm^{−1} were assigned as the stretching vibrations of -Ph, while those at 810 cm^{−1} and 1022 cm^{−1} were assigned as the stretching vibrations of Si-O-Si. The appearance and enhancement of the characteristic peak at 810 cm^{−1} indicated that a crosslinked network was generated by the hydrolysis and condensation reaction of the siloxane groups from TEOS, AF-PEG and AF-QAS [47]. The band at approximately 1724 cm^{−1} was assigned to the stretching vibration of the carbonyl group (C=O) in the carbamate (-NHCOO-). Fig. S9–S11 shows that PDMS and PMMS contained three elements: C, O, and Si, while PMGQS contained C, O, Si, N, and Cl. These results further indicated that PMGQS was successfully prepared [48]. The occurrence of elemental composition and chemical changes in PDMS, PMMS, and PMGQS film surfaces (depths of 1–5 nm) due to graft antibacterial factors of QAS were further determined by XPS spectra analysis (Fig. 1a). The peaks at 531.8, 401.9, 284.8, 196.7, 152.8 and 101.8 eV represent oxygen (O), nitrogen (N), carbon (C), chlorine (Cl), and silicon (Si), respectively [49]. The C 1 s core-level spectra of the PDMS, PMMS, and PMGQS films are shown in Fig. 1b. The binding energies of PMGQS were at approximately 286.3 and 284.8 eV, which can be ascribed to C–N and Si-C/C-H, respectively. The new peak of PMGQS at 401.9 eV is ascribed to the nitrogen group in the quaternary ammonium salt (Fig. 1c) [50,51]. Compared with PDMS and PMMS, the peak areas at C–N, C-C/C-H and N⁺ remarkably changed, suggesting that the chemical structure of PMGQS has been successfully altered.

3.2. Surface energy and mechanical properties

PDMS elastomers are renowned for their exceptional antifouling capabilities, which can be attributed to its low surface energy and elastic modulus. However, the surface energy and mechanical properties are mutually exclusive for a silicone coating. Significantly, the modified silicone coatings (PMMS, PMGS, PMQS and PMGQS) have achieved equilibrium between both properties.

Baier's curve exhibits the minimum attachment strength from 20 mJ/m² to 25 mJ/m², which is caused by forming weak boundary layers between the coating surface and the adhesives of fouling strains [39]. We studied the surface energy of silicone coatings. As shown in Fig. 2a, The modified silicone coating PMMS shows a lower surface energy of approximately 19 mJ/m² compared with the primary PDMS (ca. 22 mJ/m²). The modified silicone coatings of PMGS, PMQS and PMGQS have a surface energy of approximately 23 mJ/m², which is slightly higher than those of PDMS and PMMS due to the introduction of the hydrophilic antifouling groups as AF-PEG and AF-QAS. These findings suggest that the modified PMGS, PMQS, and PMGQS possess low surface energy, facilitating the effortless removal of weakly attached marine organisms.

We measured the tensile strength of the primary PDMS film and the other modified silicone film. As shown in Fig. 2b, with the addition of OH-MPS, the tensile strength increased from 0.7 MPa to 6.4 MPa and above for all the modified variants. The modified film of PMGQS containing 25 wt% OH-MPS, 5 wt% AF-PEG and 5 wt% AF-QAS was chosen due to its optimal comprehensive mechanical property with tensile strength (5.4 MPa) and elongation at break (242%). The tensile strength

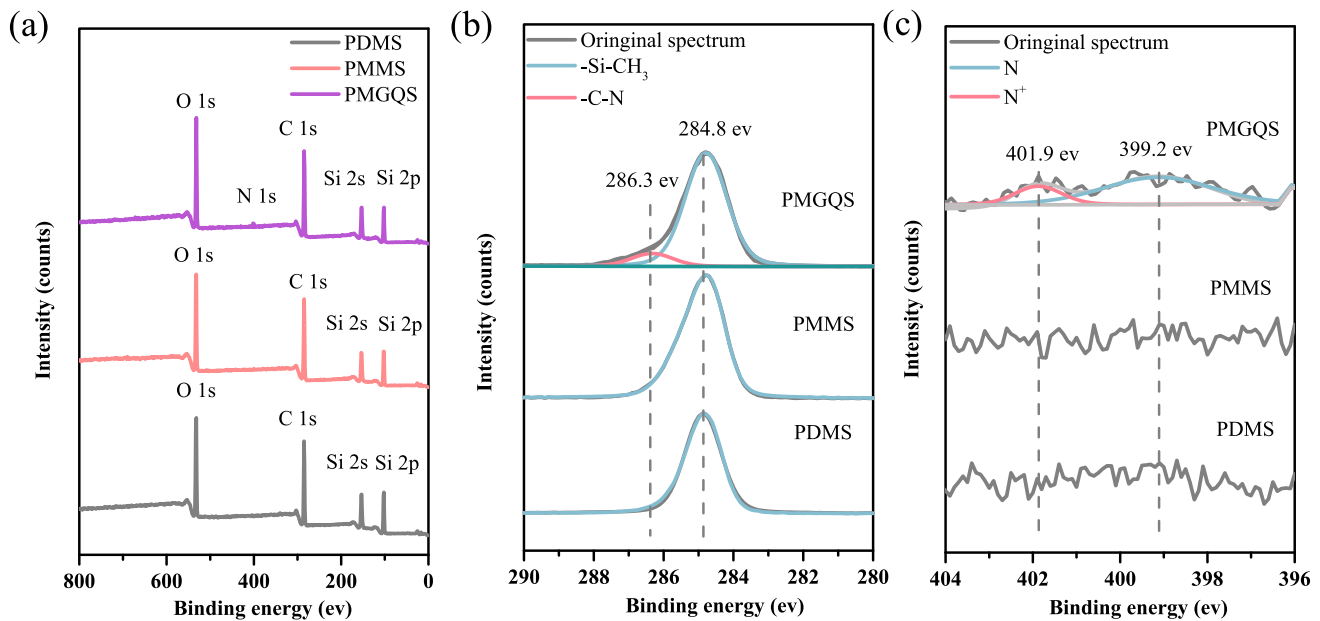


Fig. 1. Compositions of PDMS, PMMS and PMGQS polymers. (a) XPS wide-scan spectra (b) C 1 s XPS high-resolution spectra. (c) N 1 s XPS high-resolution spectra.

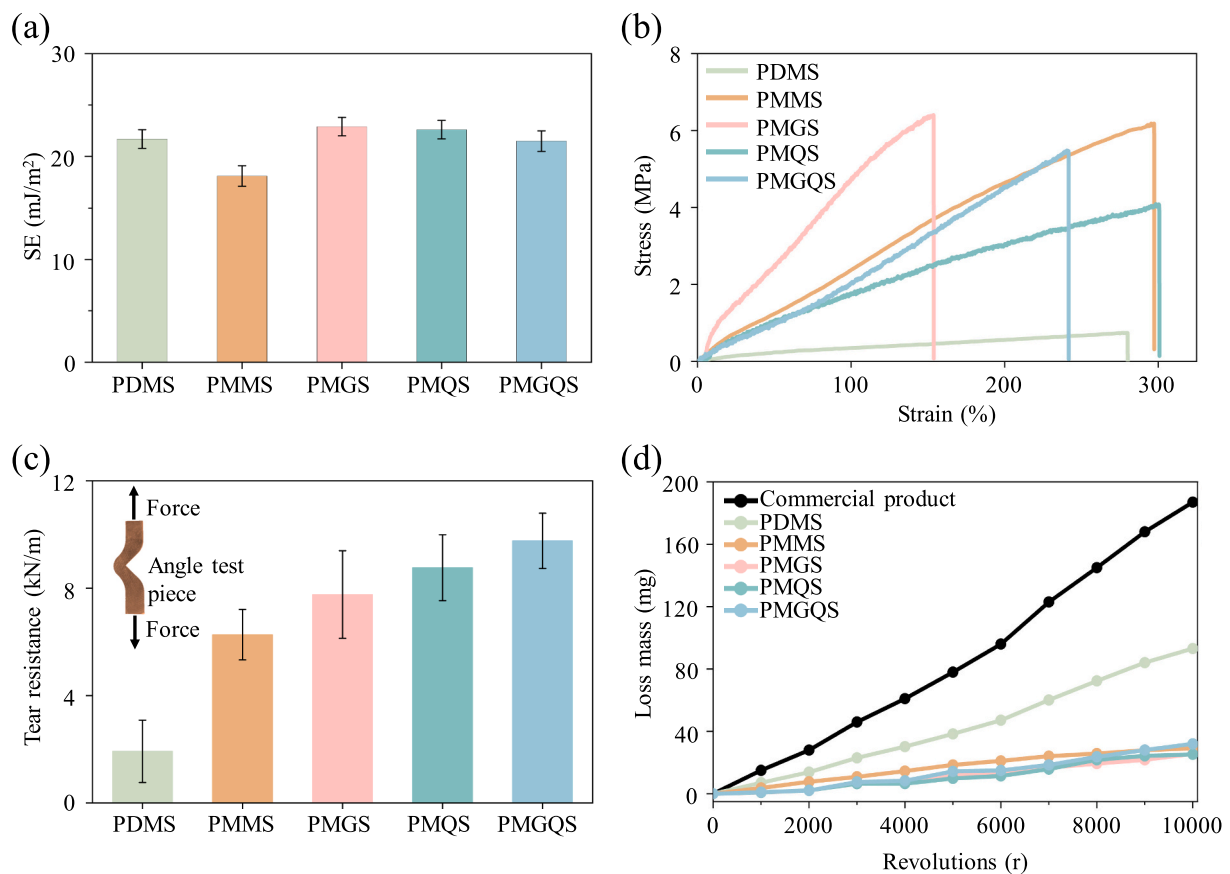


Fig. 2. Surface energy and mechanical properties. (a) Surface energy (b) The tensile stress-strain curves and (c) the corresponding shear strength of different coatings. (d) Mass loss of the different coatings after abrasion for 10,000 revolutions.

of PMGQS was approximately 7.7 times stronger than that of the primary PDMS, while its elongation to break only had a 7% decrease, still maintaining a highly ductile performance. To evaluate the anti-tearing performance, the samples were made into right-angle shapes according to ISO34-1: 2004 for tear strength measurement, The tear strength of

PMGQS was 9.8 kN/m, significantly higher than the primary PDMS from 1.9 kN/m (Fig. 2c). All the variations exhibited more than tripling enhancement in tear strength from that of PDMS, presenting a promising solution for overcoming the issues linked with low anti-scratch and anti-tearing performance of PDMS-based antifouling coatings.

To test the wear resistance, the mass loss curve of different coatings was plotted against the number of abrasion wheel rotations. As shown in Fig. 2d, the mass loss of a PDMS-based FRC commercial product was Ca. 180 mg after wearing for 10,000 rotations. In contrast, the modified

silicone coatings (PMMS, PMGS, PMQS and PMGQS) were found to have a weight loss lower than that of Ca. 30 mg, while the mass loss of the PDMS was Ca. 90 mg. Few previous studies have reported using organic coatings that could achieve <30 mg after 10,000 r wear cycles, but with

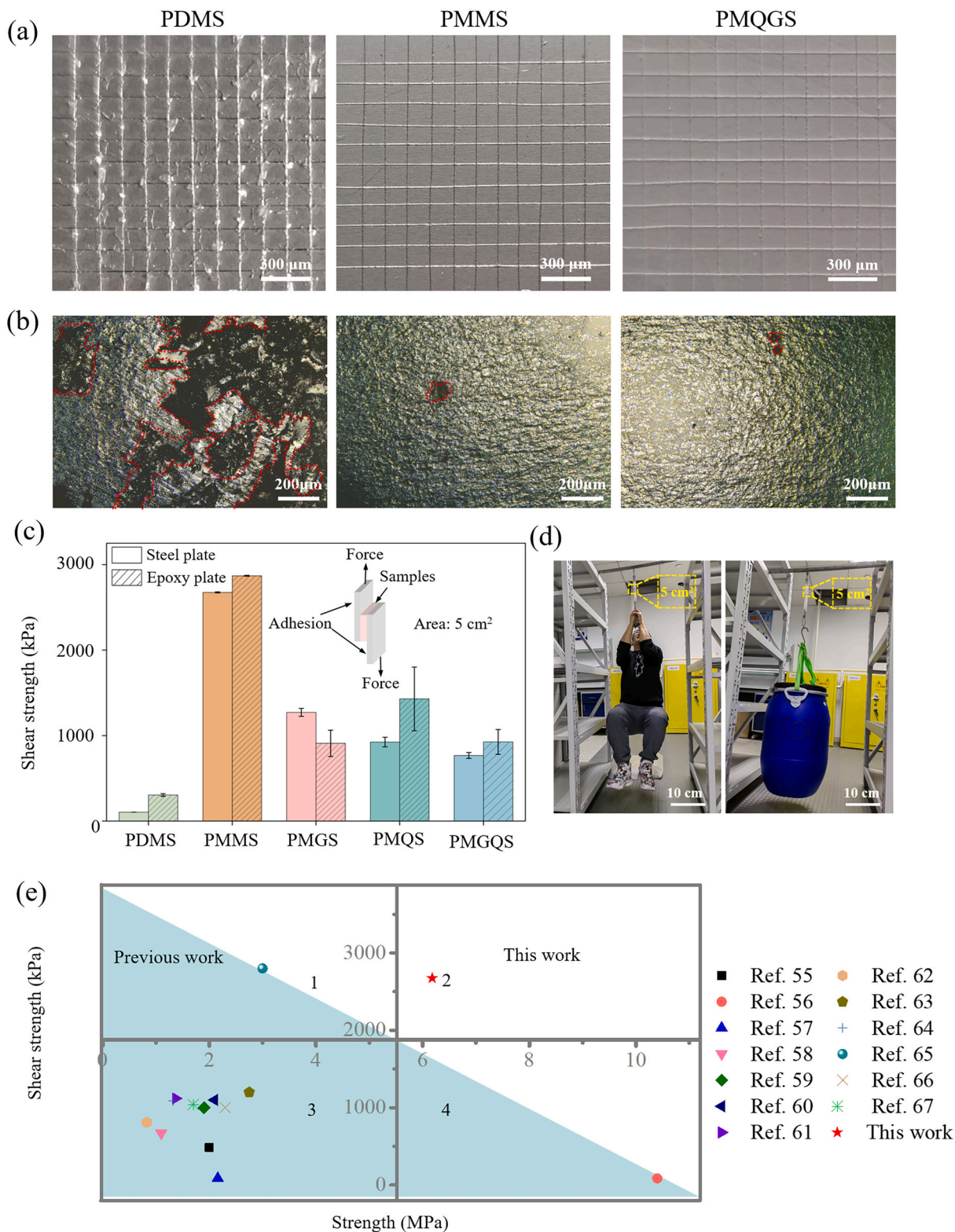


Fig. 3. (a) Cross hatch adhesion test (b) Morphology after impact resistance test (red dotted lines marking damaged areas). (c) Lap shear strength of the coatings on steel and epoxy primed plates. (d) the PMGQS adhesion strength by using a lap bonded structure to withstand a 80 kg man or a 50 kg cargo container. (e) Comparative data of strength and lap shear strength from this work to those reported in literatures: The red star is from this work, and others represent the value in reported literature [55–67].

a higher mass loss by wearing [44,52,53]. These results indicated that the modified silicone coatings (PMMS, PMGS, PMQS and PMGQS) all have excellent wear resistance due to the polymer network containing rigid segments from methyl phenyl silicon, which enhanced not only tensile strength and elongation at break but also wear resistance.

3.3. Adhesive properties

Due to their low surface energy, PDMS-based antifouling coatings are usually applied on top of the tie-coat [54]. Here, we sprayed the modified silicone coatings directly on two types of substrates (a steel plate and an epoxy primed plate) without using tie coat and measured the adhesion properties by cross hatch adhesion test, impact resistance test and the lap-shear test. In Fig. 3a, the cross hatch adhesion test results were in an excellent grade 5B of the assessment: after removing the 3 M Scotch tape from the sample surface, not a single fragment detached from the cut test area of samples PMMS and PMGQS. The edges of all the squares were clear and straight with no chipping, no detachment. On contrast, the PDMS coating showed a noticeable edge chipping and film detachment. The impact test results provide further evidence of the exceptional mechanical properties and adhesion of the PMMS and PMGQS coatings in Fig. 3b. Remarkably, minimal damage was observed after impact tests from drop heights of 50 and 100 cm, highlighting the robustness of PMMS and PMGQS. In contrast, the PDMS coating exhibited damage areas exceeding 50% even at a mere 10 cm drop height.

In addition, the adhesion strength of the primary PDMS was only approximately 104 kPa on the steel substrate and 304 kPa on the epoxy primed surface in Fig. 3c. The PMMS coating was as high as 2674 kPa and 2869 kPa on the two substrates, which was 25 times higher than that of the primary PDMS on the steel plate and approximately 9 times higher than that of primary PDMS on the epoxy primer. Although the adhesion strength decreased to some extent when AF-PEG and AF-QAS were grafted to the PMMS cross-linking system, it was still much higher than that of the primary PDMS. The adhesion strength of the PMGQS coating was 767 kPa and 925 kPa on the steel plate and epoxy primer, respectively. We further verified the adhesion strength by bonding two steel strips using PMGQS with a lap area of 2.5 cm x 2 cm. Surprisingly, the bonded strength was sufficient to withstand the weight of a 50 kg cargo container or an 80 kg lab technician in Fig. 3d (Video S1). A bonded strength of >1600 kPa could be extracted from this test. These results indicated that the PMGQS coating can provide a good adhesion, which could overcome the weakness of FRCs based on low surface energy silicone materials in industrial applications. To show the improvement of the mechanical properties and the adhesion property of modified silicone elastomers, we compared the lap shear strength as a function of stress strength of several representative synthetic silicone materials. As depicted in Fig. 3e, our PMGQS coating clearly outperforms the others in terms of strength and lap shear strength. In contrast, most modified silicone elastomers reported in literature exhibit low to moderate strength and lap shear strength, typically falling within the third quartile ^{46–52}. Moreover, there is a noticeable disparity between mechanical strength and lap shear strength for coatings situated in the first and fourth quartiles ^{47,52}. Therefore, these results show that the PMGQS displayed super strong mechanical properties and adhesion among the reported silicon elastomer variants, which presents a promising and viable solution for marine applications.

3.4. Antifouling performance

3.4.1. Antibacterial properties

When solid surface is submerged in seawater, the key to inhibiting biofouling is to prevent the colonization of bacteria on the surface [68]. QAS is an effective broad-spectrum antibacterial agent [69]. To evaluate the antibacterial property, *E. coli* (gram-negative bacteria) and *S. aureus* (gram-positive bacteria) were selected as representatives of the broad

spectrum antibacterial evaluation. Images of bacterial colony growth on different samples are shown in Fig. 4a. Compared with the blank control, a significant decrease in *E. coli* and *S. aureus* for the antibacterial-factor-modified PMGS, PMQS and PMGQS films was observed. The antibacterial ratios of PMGS, PMQS, and PMGQS against *E. coli* were 98.9%, 98.9%, and 99.4%, respectively. The antibacterial ratios of PMGS, PMQS and PMGQS against *S. aureus* were higher, at 99.4%, 99.4%, and 99.7%, respectively (Fig. 4c). They were much higher than those of PDMS and PMMS in which the antibacterial ratio for PDMS is approximately 2.2% (*E. coli*) and 27.9% (*S. aureus*) whereas for PMMS, the ratios are 20.2% and 37.7%, respectively. The low efficacy is ascribed to the lack of a fouling-resistant or antibacterial moiety on the PDMS and PMMS surface. These results illustrate that the AF-PEG or AF-QAS component endows PMGS, PMQS and PMGQS with an excellent antibacterial effect.

3.4.2. Diatom inhibition properties

Polydimethylsiloxane (PDMS) elastomers have been used as FRC anti-fouling coatings in the marine shipping industry due to their high smooth topology, water repellency and low surface free energy [9,40]. However, silicone can become the colonization site of diatoms because the hydrophobic solid surface seems to favor the adhesion of such fouling organisms [70]. To investigate the anti-fouling property, *Navicula* and *Nitzschia closterium* were used as typical diatom models to evaluate the anti-diatom adhesion resistance of the different modified silicone coatings. Fig. 4b shows the fluorescence microscopic images of *Navicula* and *Nitzschia closterium* adhered on different samples. Notably, the lowest colony of *Navicula* and *Nitzschia closterium* were observed on the surface of PMGQS, while a large colony of *Navicula* and *Nitzschia closterium* were observed on the surface of primary PDMS. As shown in Fig. 4d, the *Navicula* and *Nitzschia closterium* density of the PDMS reached 136.6 ± 2.1 and 109.2 ± 1.7 diatoms/mm², respectively. PMMS also has a high diatom density due to its higher hydrophobicity. In contrast, the diatom density was as low as 11.2 ± 1.1 and 0.5 ± 0.5 diatoms/mm² on the PMGQS. With the introduction of AF-PEG and AF-QAS, there was a notable decrease in the populations of *Navicula* and *Nitzschia closterium* difficile by 91.8% and 99.5% respectively, which served as strong evidences of PMGQS achieving optimal anti-diatom properties.

3.5. Marine field test

We further investigated the anti-fouling performance of modified silicone-based coatings in an actual marine environment for 12 months. Fig. 5a shows the photos of the coated panels through 0, 6 and 12 months immersion in Xiamen Bay (South China Sea), Fig. 5b displays the corresponding quantitative data of the area covered by biofouling. After immersion in the sea for 6 months, the primary PDMS coated panel was almost completely covered with various biospecies, and the 2nd most fouled coating was PMMS. The fouling species on these two panels consisted of tube worms, bryozoans and algae. In contrast, PMQS and PMGQS just showed little fouling after 6 months of immersion with less than ~5% fouling coverage (Fig. 5b). After immersion for 12 months, the plates coated with PMGS and PMQS were fouled with ~81% and ~55% covered area, respectively. The surface of PMGQS had only 20% of its area covered with slime only, indicating its remarkable antifouling performance under static marine condition. Interestingly, the actual marine field test results are consistent with the laboratory test results on antidiatoms. Therefore, PMGQS with integrated functions can provide a promising solution for long-term antifouling performance.

3.6. Mechanism of antifouling and enhanced mechanical properties of PMGQS

A commonly used approach to fulfill antibacteria and antifouling functions hydrophilic molecules such as quaternary ammonium salt and polyethylene glycol were physically blended into a coating tend to

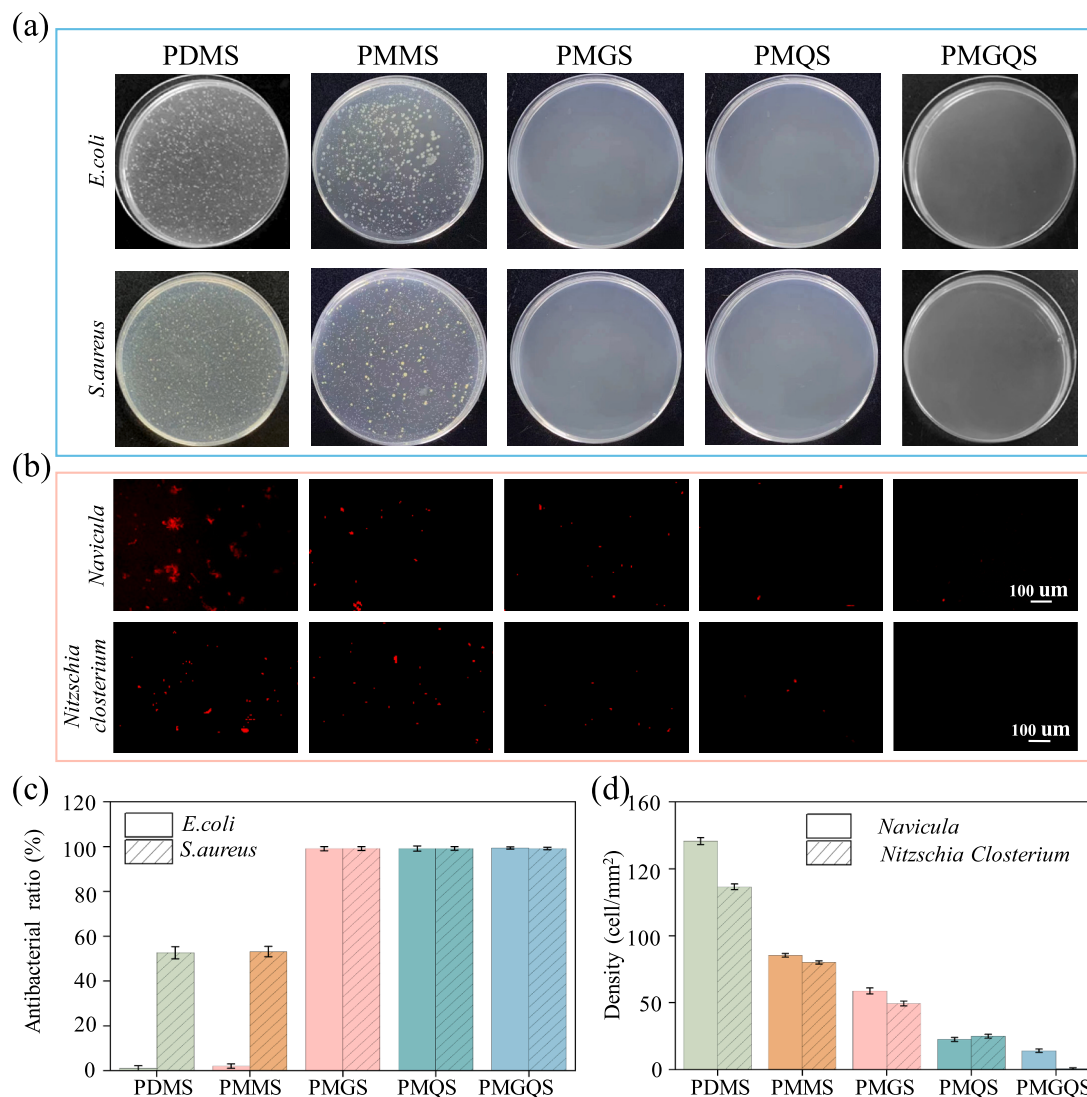


Fig. 4. Anti-biofouling performance. (a) Antibacterial images of the silicone-based anti-fouling coatings for *E. coli* and *S. aureus*. (b) Fluorescence microscopy images of *Navicula* and *Nitzschia closterium* attached to different coatings for 24 h; the scale bar is 100 μm . (c) Comparison of the antibacterial ratio of different samples. (d) Quantitative analysis showing the density of diatoms attached to different coatings.

diffuse or dissolve in water [71]. With the aim to achieve a sustainable performance by preventing the release of these two components in seawater, both AF-QAS and AF-PEG were grafted into the PDMS cross-linking structure. Thus they can be anchored on the surface to form a top layer instead of being dissolved in seawater, possessing enriched antifouling functional groups (Fig. 6a). This anchoring strategy, provides not only the antibacterial efficiency, but also a long-term efficacy. The surface wettability changes of the primary PDMS, PMMS, PMGS, PMQS and PMGQS were evaluated by measuring the static water contact angle (WCA) on the coating surface before and after being immersed in an artificial seawater (ASW) for 7 days. As shown in Fig. S13, it is interesting to note that the WCAs of the modified PMGS, PMQS and PMGQS films decreased by 4°, 6° and 3°, respectively. In contrast, the primary PDMS and PMMS films increased by 1° after soaking for 7 days. This phenomenon indicated that the AF-QAS and AF-PEG groups grafted to the PDMS cross-linking system mobilized to the coating surface during immersion and reduced WCAs due to their hydrophilic characteristics. To further prove that the hydrophilic AF-QAS was enriched on the coating surface after soaking in the PMGQS sample, EDS spectra from SEM of the coating section were obtained. As shown in Fig. S13, the weight content of characteristic elements such as nitrogen and chlorine from the quaternary ammonium salt group increased from 1.99% to

2.66% and 0.16% to 0.21% after soaking for 7 days, respectively, indicating that the hydrophilic groups migrated to the coating surface conclusively. With the occurrence of composition self-assembly, the surface morphology of the coating was changed accordingly. We explored the surface morphology of these coatings by LSCM and AFM (Fig. S14 and S15). After immersion for 14 days, the surface roughness of the PMGQS film increased from $\sim 0.1 \mu\text{m}$ to $\sim 5.0 \mu\text{m}$, while the primary PDMS remained almost unchanged. These results suggested that the functional groups of QAS and PEG were migrated to the surface and anchored on it to play a role of fouling protection in seawater.

The enhanced mechanical property of the PMGQS coatings can be attributed to the synergies of hydrogen bonding, π - π aromatic interactions, and covalent bonding complexation. Hydrogen bonding and covalent bonding come from amino ester bonds in AF-PEG and cross-linked polysiloxane, respectively. PMGQS is incorporated with a benzene ring structure to form a micro-phase partition (separation) which can contribute to a substantial enhancement in the mechanical strength of the coatings. In addition, the presence of silanol segments in the coatings fosters robust adhesion to the substrate through hydrogen bonding and covalent bonding, as illustrated in Fig. 6a. These synergies contribute to the development of coatings with exceptional mechanical properties and adhesion. The the PDMS, PMMS and PMGQS elastomers

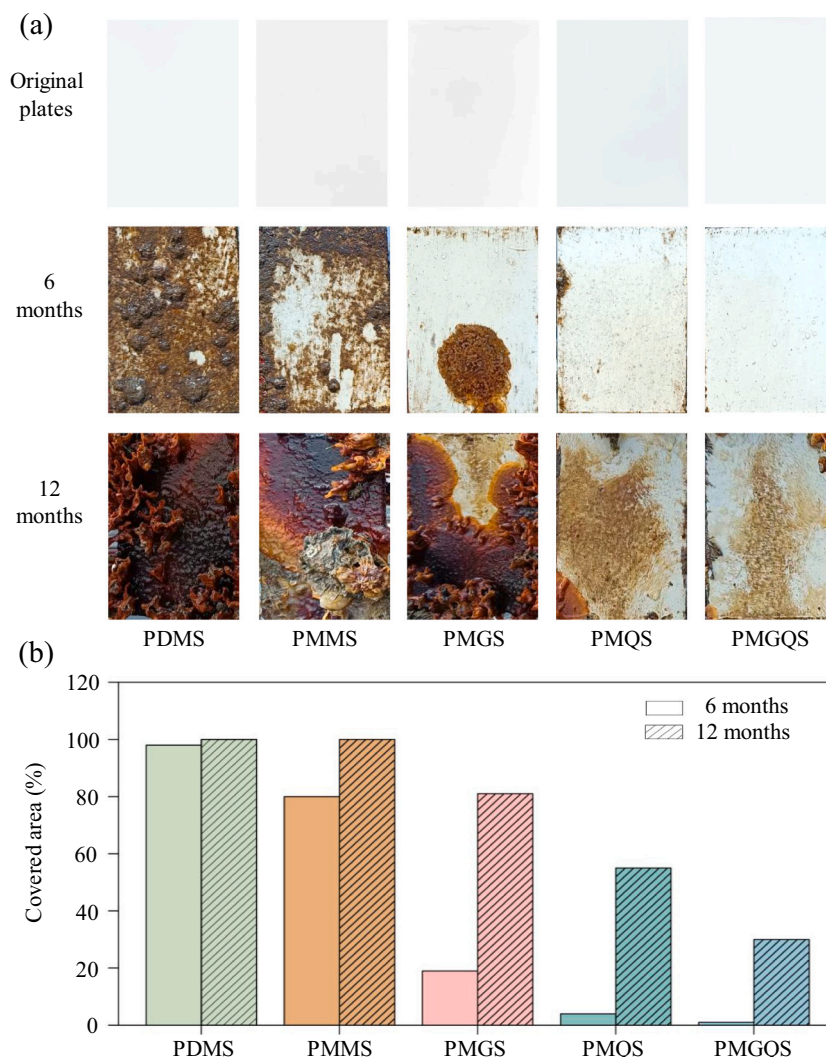


Fig. 5. (a) Images of coatings after immersion in the Xiamen Bay of China for 0–12 months (May 2021 to May 2022, Xiamen Bay). (b) Biofouling coverage area of PDMS, PMMS, PMGS, PMQS and PMGQS coatings after the 6 and 12 months immersion.

was characterized by wide-angle X-ray diffraction (WAXD) and small-angle X-ray scattering (SAXS). The WAXD patterns exhibit two broad diffraction peaks centered at $2\theta = 20^\circ$, and 12° , which correspond to the 110 plane in MPS crystals and the amorphous PDMS segments, respectively (Fig. 6b). The broad endothermic transition peak further at -42.5°C in the differential scanning calorimeter curve is assigned to the melting of MPS crystals (Fig. S16), which further supports the existence of MPS crystals in the PMGQS elastomer. The SAXS Patterns of the PMGQS elastomers in Fig. 6c reveal a prominent principal scattering peak (q^*), indicating the presence of phase-separated domains in the PMGQS. The calculated periodicity (D) of these phase-separated domains is 35.6 nm ($D = 2\pi / q^*$). Given the 75:25 ratio of PDMS to MPS in PMGQS elastomers, we postulate that dynamic interactions, including π - π aromatic interactions, are confined within crystallized PMGQS, leading to the formation of phase-separated domains with dynamic hierarchical structures. Additionally, the scattering circles in the 2D-SAXS scattering patterns (inset image in Fig. 6c) confirm the presence of electron-rich components within PMGQS crystals.

4. Conclusions

In summary, we have prepared a novel mechanically robust multifunctional silicone-based fouling release polymer with eco-friendly features by grafting OH-MPS, AF-QAS and AF-PEG to bis-silanol terminated

OH-PDMS. FTIR, SEM and XPS were used to investigate the changes in the chemical compositions, which are supporting evidences of the successful grafting of the antifoulant groups to the polymer network to provide sustainable antifouling performance. Among the variants of silicon-based polymers, the PMGQS polymer-based coating showed lower surface energy and significantly improved mechanical property, ultra strong adhesion, outstanding wear resistance and impact resistance, offering values of mechanical strength up to 6.4 MPa (9.1 times that of PDMS) and substrate adhesion up to 2.87 MPa (25.7 times that of PDMS), respectively. These improved properties of the PMGQS coating can be attributed to the synergies of hydrogen bonding, π - π aromatic interactions and covalent bonding complexation, which form a micro-phase partition.

The PMGQS polymer-based coating exhibited the highest efficacy of antibacterial activity as well as diatom inhibition and antifouling marine field test, owing to the enriched antifoulant groups restricted to the surface layer. The PMGQS coating subject to 12 months sea testing demonstrated an efficient fouling protection with 20% coverage of slime only compared to the conventional PDMS film with 100% coverage by tube worms, bryozoans, and algae. In conclusion, PMGQS provides not only much improved mechanical properties, but also sustainable and efficient antifouling performance over the conventional PDMS based coating.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.smt.2024.101015>.

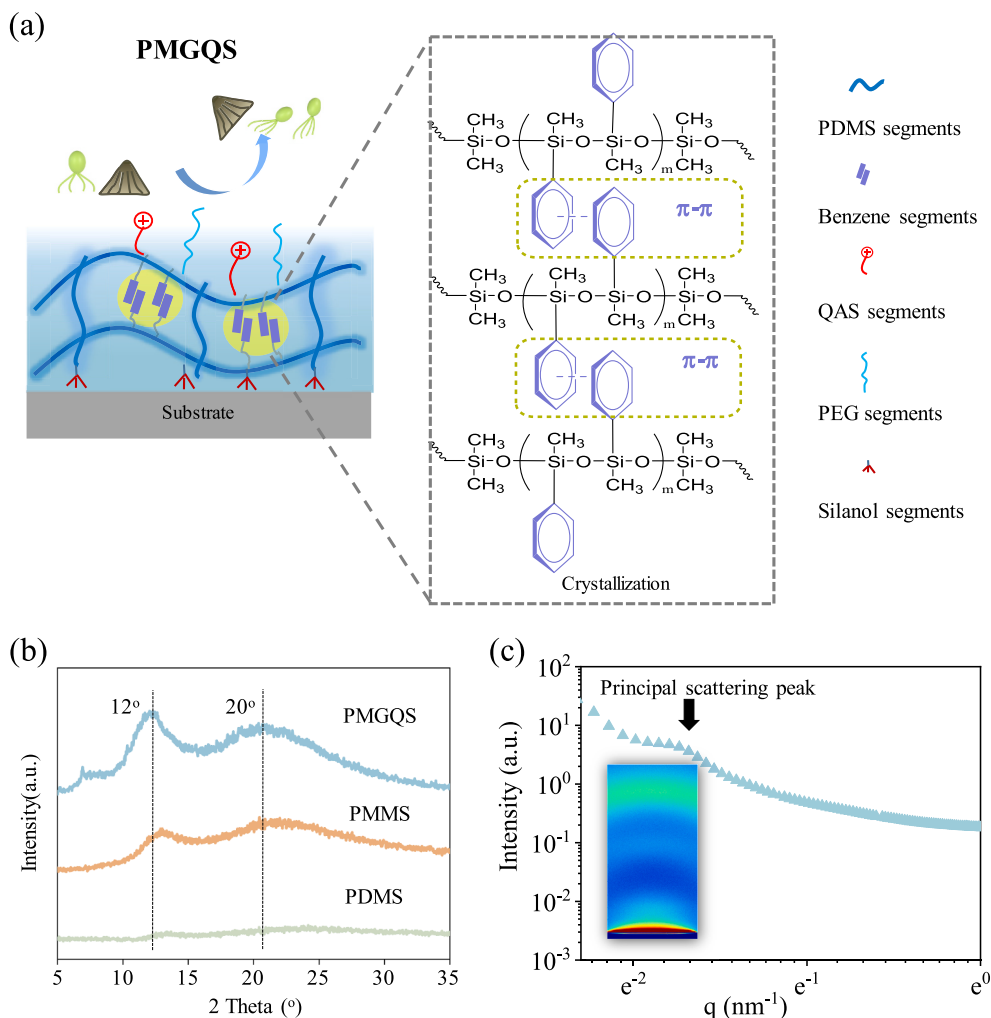


Fig. 6. (a) Mechanism of antifouling and enhanced mechanical properties of the PMGQS. (b) WAXD pattern of PDMS, PMMS and PMGQS. (c) SAXS pattern of the PMGQS elastomer. The inset shows the 2D-SAXS image of the elastomer.

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CRediT authorship contribution statement

Xiaowei An: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jiannan Cheng:** Methodology, Investigation, Formal analysis, Data curation. **Qi Li:** Writing – review & editing, Software. **Jiyun Tang:** Methodology, Data curation. **Zongkui Li:** Methodology, Data curation. **Liwei Liu:** Writing – review & editing, Funding acquisition, Conceptualization. **Hui Yang:** Writing – review & editing, Supervision, Investigation, Funding acquisition. **Chunyang Wei:** Writing – review & editing, Visualization, Validation, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

None.

Data availability

Data will be made available on request.

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